

1. (Question 1). Let X be a discrete random variable with the given probability mass function with $0 \leq \theta \leq 1$ as a parameter:

x	0	1	2	3
$P(X = x \theta)$	$2\theta/3$	$\theta/3$	$2(1 - \theta)/3$	$(1 - \theta)/3$

We have the following i.i.d. sample $\mathbf{x} = (3, 0, 2, 1, 3, 2, 1, 0, 2, 1)$ from X and we want the maximum likelihood estimate of θ . Let x^t denote the t -th observation, for $t = 1, \dots, N$, where here $N = 10$, so that $\mathbf{x} = (x^1, \dots, x^N)$. Since the observations are independent, the likelihood of the parameter θ is

$$\begin{aligned}
 l(\theta | \mathbf{x}) &= \prod_{t=1}^N p(x^t | \theta) \\
 &= p(3 | \theta) \cdot p(0 | \theta) \cdot p(2 | \theta) \cdot p(1 | \theta) \cdot p(3 | \theta) \\
 &\quad \cdot p(2 | \theta) \cdot p(1 | \theta) \cdot p(0 | \theta) \cdot p(2 | \theta) \cdot p(1 | \theta) \\
 &= \frac{1 - \theta}{3} \cdot \frac{2\theta}{3} \cdot \frac{2(1 - \theta)}{3} \cdot \frac{\theta}{3} \cdot \frac{1 - \theta}{3} \\
 &\quad \cdot \frac{2(1 - \theta)}{3} \cdot \frac{\theta}{3} \cdot \frac{2\theta}{3} \cdot \frac{2(1 - \theta)}{3} \cdot \frac{\theta}{3} \\
 &= \left(\frac{2\theta}{3}\right)^2 \left(\frac{\theta}{3}\right)^3 \left(\frac{2(1 - \theta)}{3}\right)^3 \left(\frac{1 - \theta}{3}\right)^2 \\
 &= \frac{2^5}{3^{10}} \theta^5 (1 - \theta)^5.
 \end{aligned}$$

We can optimize via the log likelihood and our desired MLE for θ is $\hat{\theta} = \arg \max_{\theta} (L(\theta | \mathbf{x}))$ so consider

$$\begin{aligned}
 L(\theta | \mathbf{x}) &= \log(l(\theta | \mathbf{x})) \\
 &= \log\left(\frac{2^5}{3^{10}} \theta^5 (1 - \theta)^5\right).
 \end{aligned}$$

Applying rules of logarithms we have

$$L(\theta | \mathbf{x}) = \log\left(\frac{2^5}{3^{10}}\right) + 5 \log \theta + 5 \log(1 - \theta).$$

Taking the derivative of L with respect to θ

$$\begin{aligned}\frac{dL}{d\theta} &= 0 + 5 \cdot \frac{1}{\theta} + 5 \cdot \frac{1}{1-\theta} \cdot (-1) \\ &= \frac{5}{\theta} - \frac{5}{1-\theta}.\end{aligned}$$

Set the derivative equal to zero and solve for θ :

$$\begin{aligned}0 &= \frac{5}{\theta} - \frac{5}{1-\theta} \\ \frac{5}{1-\theta} &= \frac{5}{\theta} \\ \theta &= 1-\theta \\ \theta &= \frac{1}{2}.\end{aligned}$$

To verify that this gives a maximum, rewrite the derivative with a common denominator:

$$\begin{aligned}\frac{dL}{d\theta} &= \frac{5}{\theta} - \frac{5}{1-\theta} \\ &= \frac{5(1-\theta) - 5\theta}{\theta(1-\theta)} \\ &= \frac{5(1-2\theta)}{\theta(1-\theta)}.\end{aligned}$$

For $0 < \theta < 1$, the denominator is positive, so the sign of the derivative is determined by $1 - 2\theta$. Thus the derivative is positive for $0 < \theta < \frac{1}{2}$, zero at $\theta = \frac{1}{2}$, and negative for $\frac{1}{2} < \theta < 1$. The log likelihood therefore increases up to $\theta = \frac{1}{2}$ and decreases afterward. Since the logarithm is strictly increasing, this also maximizes the likelihood on $(0, 1)$. At the endpoints $\theta = 0$ and $\theta = 1$, the likelihood is zero, so neither endpoint gives a larger value.

Therefore the maximum likelihood estimate of θ is $\hat{\theta} = \boxed{\frac{1}{2}}$.

2. (*Question 2*). Let $X_1, X_2, \dots, X_N$ be i.i.d. random variables with density function

$$f(x \mid \sigma) = \frac{1}{2\sigma} e^{-\frac{|x|}{\sigma}}.$$

Again, since the observations are independent, we define the likelihood function as

$$\begin{aligned}
l(\sigma \mid x) &\equiv p(x \mid \sigma) = \prod_{t=1}^N p(x_t \mid \sigma) \\
&= \prod_{t=1}^N f(x_t \mid \sigma) \\
&= \prod_{t=1}^N \left(\frac{1}{2\sigma} e^{-\frac{|x_t|}{\sigma}} \right) \\
&= \left(\frac{1}{2\sigma} \right)^N \prod_{t=1}^N \left(e^{-\frac{|x_t|}{\sigma}} \right) \\
&= \left(\frac{1}{2\sigma} \right)^N e^{\left(\sum_{t=1}^N -\frac{|x_t|}{\sigma} \right)} \\
&= \left(\frac{1}{2\sigma} \right)^N e^{\left(-\frac{1}{\sigma} \sum_{t=1}^N |x_t| \right)}.
\end{aligned}$$

We then take the log, which we will optimize.

$$\begin{aligned}
L(\sigma \mid x) &= \log(l(\sigma \mid x)) \\
&= \log \left[\left(\frac{1}{2\sigma} \right)^N e^{-\frac{1}{\sigma} \sum_{t=1}^N |x_t|} \right].
\end{aligned}$$

Applying rules of logarithms we have

$$\begin{aligned}
L(\sigma \mid x) &= \log \left[\left(\frac{1}{2\sigma} \right)^N \right] + \log \left[e^{-\frac{1}{\sigma} \sum_{t=1}^N |x_t|} \right] \\
&= N \log \left(\frac{1}{2\sigma} \right) - \frac{1}{\sigma} \sum_{t=1}^N |x_t| \\
&= -N \log(2\sigma) - \frac{1}{\sigma} \sum_{t=1}^N |x_t| \\
&= -N \log 2 - N \log \sigma - \frac{1}{\sigma} \sum_{t=1}^N |x_t|.
\end{aligned}$$

Taking the derivative of L with respect to σ

$$\begin{aligned}
\frac{dL}{d\sigma} &= 0 - N \cdot \frac{1}{\sigma} - \left(-\frac{1}{\sigma^2}\right) \sum_{t=1}^N |x_t| \\
&= -\frac{N}{\sigma} + \frac{1}{\sigma^2} \sum_{t=1}^N |x_t|.
\end{aligned}$$

Set the derivative equal to zero and solve for σ :

$$\begin{aligned}
0 &= -\frac{N}{\sigma} + \frac{1}{\sigma^2} \sum_{t=1}^N |x_t| \\
\frac{N}{\sigma} &= \frac{1}{\sigma^2} \sum_{t=1}^N |x_t|
\end{aligned}$$

Since $\sigma > 0$, we have $\sigma^2 > 0$, so we can multiply both sides by σ^2 :

$$\begin{aligned}
N\sigma &= \sum_{t=1}^N |x_t| \\
\sigma &= \frac{1}{N} \sum_{t=1}^N |x_t|.
\end{aligned}$$

To verify that this gives a maximum, let $\hat{\sigma} = \frac{1}{N} \sum_{t=1}^N |x_t|$ and rewrite the derivative with a common denominator:

$$\begin{aligned}
\frac{dL}{d\sigma} &= -\frac{N}{\sigma} + \frac{1}{\sigma^2} \sum_{t=1}^N |x_t| \\
&= \frac{\sum_{t=1}^N |x_t| - N\sigma}{\sigma^2} \\
&= \frac{N\hat{\sigma} - N\sigma}{\sigma^2} \\
&= \frac{N(\hat{\sigma} - \sigma)}{\sigma^2}.
\end{aligned}$$

For $\sigma > 0$, the factor N/σ^2 is positive, so the sign of the derivative is determined by $\hat{\sigma} - \sigma$. When $\sum_{t=1}^N |x_t| > 0$, we have $\hat{\sigma} > 0$. Thus the derivative is positive for $0 < \sigma < \hat{\sigma}$, zero at $\sigma = \hat{\sigma}$, and negative for $\sigma > \hat{\sigma}$. The log likelihood therefore increases up to $\hat{\sigma}$ and decreases afterward, so this gives the maximum. Since the logarithm is strictly increasing, maximizing the likelihood is equivalent to maximizing its log. Therefore the maximum likelihood estimate of σ is

$$\begin{aligned}
\hat{\sigma} &= \arg \max_{\sigma > 0} l(x \mid \sigma) \\
&= \arg \max_{\sigma > 0} \log(l(x \mid \sigma)) \\
&= \arg \max_{\sigma > 0} L(\sigma \mid x) \\
&= \boxed{\frac{1}{N} \sum_{t=1}^N |x_t|}.
\end{aligned}$$

3. (*Question 3*). We are given N random samples $X_1, X_2, \dots, X_N$ where we assume that the density function is the Gaussian:

$$f(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)}$$

We use MLE to estimate the parameters μ and σ . I prefer e notation, but the exponents are getting very large for this problem so I am going to switch to $\exp$:

$$\begin{aligned}
l(\mu, \sigma \mid x) &\equiv p(x \mid \mu, \sigma) = \prod_{t=1}^N p(x_t \mid \mu, \sigma) \\
&= \prod_{t=1}^N f(x_t \mid \mu, \sigma^2) \\
&= \prod_{t=1}^N \left[\frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x_t - \mu)^2}{2\sigma^2}\right) \right] \\
&= \left(\frac{1}{\sqrt{2\pi}\sigma}\right)^N \prod_{t=1}^N \exp\left(-\frac{(x_t - \mu)^2}{2\sigma^2}\right) \\
&= \left(\frac{1}{\sqrt{2\pi}\sigma}\right)^N \exp\left[\sum_{t=1}^N \left(-\frac{(x_t - \mu)^2}{2\sigma^2}\right)\right] \\
&= \frac{1}{(2\pi)^{N/2}\sigma^N} \exp\left[-\frac{1}{2\sigma^2} \sum_{t=1}^N (x_t - \mu)^2\right].
\end{aligned}$$

This is old hat at this point, we take the log to make differentiation easier.

$$\begin{aligned}
L(\mu, \sigma \mid x) &= \log(l(\mu, \sigma \mid x)) \\
&= \log \left[\frac{1}{(2\pi)^{N/2} \sigma^N} \exp \left(-\frac{1}{2\sigma^2} \sum_{t=1}^N (x_t - \mu)^2 \right) \right] \\
&= \log \left(\frac{1}{(2\pi)^{N/2} \sigma^N} \right) + \log \left[\exp \left(-\frac{1}{2\sigma^2} \sum_{t=1}^N (x_t - \mu)^2 \right) \right] \\
&= -\log((2\pi)^{N/2} \sigma^N) - \frac{1}{2\sigma^2} \sum_{t=1}^N (x_t - \mu)^2 \\
&= -\log((2\pi)^{N/2}) - \log(\sigma^N) - \frac{1}{2\sigma^2} \sum_{t=1}^N (x_t - \mu)^2 \\
&= -\frac{N}{2} \log(2\pi) - N \log \sigma - \frac{1}{2\sigma^2} \sum_{t=1}^N (x_t - \mu)^2.
\end{aligned}$$

Since we are estimating for both μ and σ and L is a function of both, we will have to take partial derivatives. First with respect to μ :

$$\begin{aligned}
\frac{\partial L}{\partial \mu} &= 0 - 0 - \frac{1}{2\sigma^2} \sum_{t=1}^N 2(x_t - \mu) \cdot (-1) \\
&= \frac{1}{\sigma^2} \sum_{t=1}^N (x_t - \mu) \\
&= \frac{1}{\sigma^2} \left(\sum_{t=1}^N x_t - N\mu \right).
\end{aligned}$$

Then with respect to σ :

$$\begin{aligned}
\frac{\partial L}{\partial \sigma} &= 0 - N \cdot \frac{1}{\sigma} - \frac{1}{2} \cdot (-2\sigma^{-3}) \sum_{t=1}^N (x_t - \mu)^2 \\
&= -\frac{N}{\sigma} + \frac{1}{\sigma^3} \sum_{t=1}^N (x_t - \mu)^2.
\end{aligned}$$

Now we set these partial derivatives equal to 0 and solve.

$$0 = \frac{1}{\sigma^2} \left(\sum_{t=1}^N x_t - N\mu \right)$$

Since $\sigma > 0$, we can multiply both sides by σ^2 :

$$\begin{aligned} 0 &= \sum_{t=1}^N x_t - N\mu \\ N\mu &= \sum_{t=1}^N x_t \\ \mu &= \boxed{\frac{1}{N} \sum_{t=1}^N x_t = \bar{x}}. \end{aligned}$$

For σ , substituting $\mu = \bar{x}$ gives

$$\begin{aligned} 0 &= -\frac{N}{\sigma} + \frac{1}{\sigma^3} \sum_{t=1}^N (x_t - \bar{x})^2 \\ \frac{N}{\sigma} &= \frac{1}{\sigma^3} \sum_{t=1}^N (x_t - \bar{x})^2 \end{aligned}$$

Since $\sigma > 0$, we can multiply both sides by σ^3 :

$$\begin{aligned} N\sigma^2 &= \sum_{t=1}^N (x_t - \bar{x})^2 \\ \sigma^2 &= \frac{1}{N} \sum_{t=1}^N (x_t - \bar{x})^2 \end{aligned}$$

Taking the positive square root, provided the samples are not all identical:

$$\sigma = \boxed{\sqrt{\frac{1}{N} \sum_{t=1}^N (x_t - \bar{x})^2}}.$$

These are, of course, the familiar formulas when working with the normal distribution from statistics.

4. (*Question 4*). Let $X_1, X_2, \dots, X_N$ be an i.i.d. sample from a Poisson distribution with parameter λ and probability mass function

$$P(X = x \mid \lambda) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x = 0, 1, 2, \dots$$

We use MLE to estimate λ . Again, since the observations are independent, the likelihood is

$$\begin{aligned}
l(\lambda \mid x) &= \prod_{t=1}^N p(x_t \mid \lambda) \\
&= \prod_{t=1}^N \frac{\lambda^{x_t} e^{-\lambda}}{x_t!} \\
&= \frac{\left(\prod_{t=1}^N \lambda^{x_t} \right) \left(\prod_{t=1}^N e^{-\lambda} \right)}{\prod_{t=1}^N x_t!} \\
&= \frac{\lambda^{\sum_{t=1}^N x_t} e^{-N\lambda}}{\prod_{t=1}^N x_t!}.
\end{aligned}$$

We then take the log to make differentiation easier. For $\lambda > 0$, applying rules of logarithms gives

$$\begin{aligned}
L(\lambda \mid x) &= \log(l(\lambda \mid x)) \\
&= \log\left(\frac{\lambda^{\sum_{t=1}^N x_t} e^{-N\lambda}}{\prod_{t=1}^N x_t!}\right) \\
&= \log\left(\lambda^{\sum_{t=1}^N x_t}\right) + \log(e^{-N\lambda}) - \log\left(\prod_{t=1}^N x_t!\right) \\
&= \left(\sum_{t=1}^N x_t\right) \log \lambda - N\lambda - \sum_{t=1}^N \log(x_t!).
\end{aligned}$$

Taking the derivative of L with respect to λ

$$\begin{aligned}
\frac{dL}{d\lambda} &= \left(\sum_{t=1}^N x_t\right) \cdot \frac{1}{\lambda} - N - 0 \\
&= \frac{1}{\lambda} \sum_{t=1}^N x_t - N.
\end{aligned}$$

Set the derivative equal to zero and solve for λ :

$$\begin{aligned}
0 &= \frac{1}{\lambda} \sum_{t=1}^N x_t - N \\
N &= \frac{1}{\lambda} \sum_{t=1}^N x_t
\end{aligned}$$

Since $\lambda > 0$, we can multiply both sides by λ :

$$N\lambda = \sum_{t=1}^N x_t$$

$$\lambda = \frac{1}{N} \sum_{t=1}^N x_t = \bar{x}.$$

To verify that this gives a maximum when $\bar{x} > 0$, rewrite the derivative as

$$\begin{aligned} \frac{dL}{d\lambda} &= \frac{\sum_{t=1}^N x_t - N\lambda}{\lambda} \\ &= \frac{N(\bar{x} - \lambda)}{\lambda}. \end{aligned}$$

For $\lambda > 0$, the factor N/λ is positive, so the sign of the derivative is determined by $\bar{x} - \lambda$. Thus the derivative is positive for $0 < \lambda < \bar{x}$, zero at $\lambda = \bar{x}$, and negative for $\lambda > \bar{x}$. The log likelihood therefore increases up to $\bar{x}$ and decreases afterward. Since the logarithm is strictly increasing, this also maximizes the likelihood.

Therefore, on the parameter space $\lambda > 0$, the maximum likelihood estimate is

$$\begin{aligned} \hat{\lambda} &= \arg \max_{\lambda > 0} l(\lambda \mid x) \\ &= \arg \max_{\lambda > 0} L(\lambda \mid x) \\ &= \boxed{\frac{1}{N} \sum_{t=1}^N x_t = \bar{x}}. \end{aligned}$$